Knowing When Not to Answer: A Multi-Domain Benchmark and Lightweight Alignment-Based Abstention for Medical VQA

Anonymous ACL submission

Abstract

Vision-language models such as LLaVA-Med perform well on medical VQA but cannot recognize unanswerable questions, a critical gap in clinical settings. We address this with a purpose-built dataset of answerable and unanswerable pairs across five imaging modalities, generating unanswerable examples from LLaVA-Med's own training corpus by selecting visually mismatched but linguistically similar statements with BioMedCLIP and Qwen2.5. We then inject BioMedCLIP correlation vectors, the element-wise product of normalized image and text embeddings, into a frozen LLaVA-Med through a single linear projection layer. A jointly trained layer raises correct abstention from under 15% to 87.5–93.9% on four of the five modalities, at the cost of slightly higher abstention on answerable questions. Single-modality layers transfer poorly, which we trace to modality-specific structure in the alignment signal. General-purpose VLMs answer more accurately but abstain correctly on at most 49.2% of unanswerable questions, showing that answer accuracy and answerability awareness are distinct. Our code and dataset are available on GitHub.

1 Introduction

Visual Question Answering (VQA) has evolved from conventional formulations in which a model jointly interprets an image and a natural-language question to predict an answer, to increasingly sophisticated settings involving visual grounding, compositional reasoning, multi-step reasoning and robustness. (Antol et al., 2015; Zhu et al., 2016; Goyal et al., 2017; Krishna et al., 2017; Anderson et al., 2018; Tan and Bansal, 2019; Hudson and Manning, 2019). LLaVA (Liu et al., 2023a) introduced visual instruction tuning to enable large language models to follow multimodal instructions grounded in images, providing a general framework for conversational vision-language understanding.

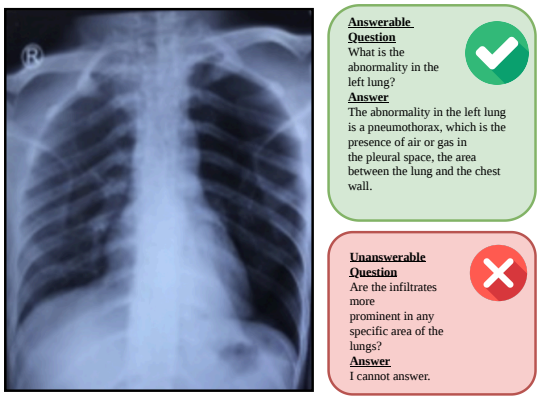


Figure 1: Illustration of image-grounded answerable and unanswerable questions in medical VQA.

Building on this framework, LLaVA-Med (Li et al., 2023a) adapted LLaVA to the biomedical domain through biomedical concept alignment and medical instruction tuning, enabling open-ended biomedical visual conversation and medical VQA. Since its introduction, LLaVA-Med has been adopted across a range of biomedical vision-language applications (Lu et al., 2024; Chen et al., 2025; Yang et al., 2025; Guo and Terzopoulos, 2024; Jiang et al., 2025).

Although current VLMs achieve strong performance on a wide range of visual-language tasks, a key limitation is that their training data are largely centered on valid, answerable interactions, which can encourage models to respond even when a question is incompatible with the image (Kafle et al., 2019; Davis, 2020; Prabhakar et al., 2018). Similarly, LLaVA-Med is trained on biomedical image-caption pairs and instruction-following conversations to support medical visual dialogue and VQA, but its original training does not explicitly model question answerability or abstention. As illustrated in Fig. 1, an image may support some questions while leaving others unanswerable due to insufficient visual evidence. LLaVA-Med was trained using biomedical image–text pairs

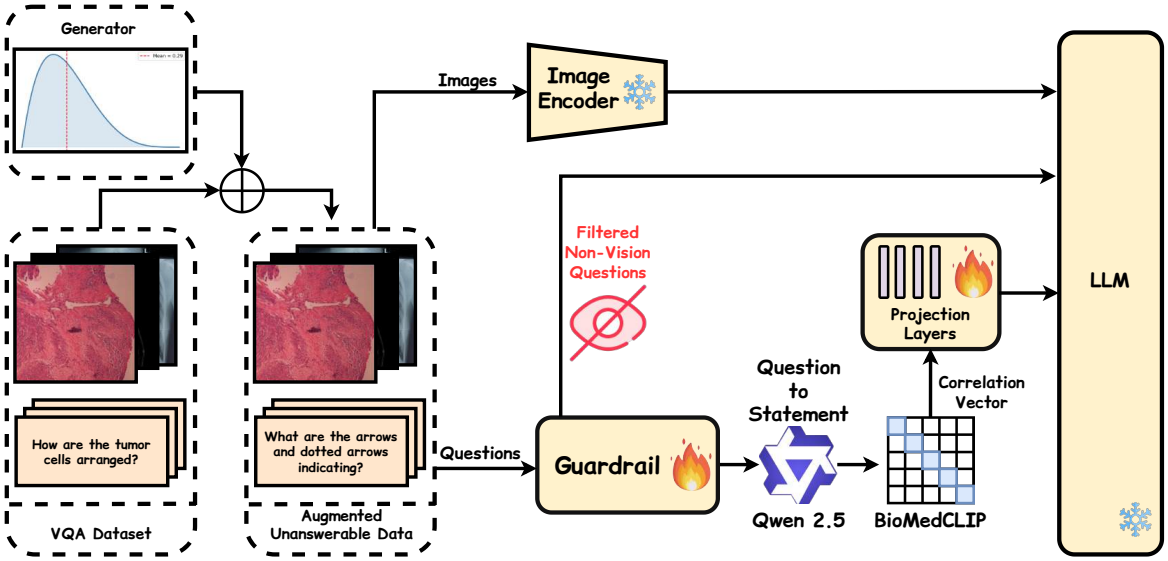


Figure 2: Overview of the proposed medical answerability pipeline. Unanswerable image–question pairs are generated from the original VQA dataset, while non-visual questions are filtered using a guardrail. The remaining questions are converted into statements using Qwen 2.5 and aligned with their images using BioMedCLIP. The resulting correlation representation is passed through specialized projection layers and provided to the LLM to support answerability-aware response generation.

from PMC-15M (Zhang et al., 2023a), and GPT-4 (OpenAI, 2022, 2023) was used to convert these captions and associated context into biomedical instruction-following conversations. However, this data-generation process was designed to produce valid image-grounded questions and responses, not explicitly unanswerable questions or abstention examples, leaving answerability detection outside the original training objective. Thus, despite their strong performance, these models cannot yet be considered fully reliable, as they remain susceptible to hallucinations and may generate confident responses even when the available visual evidence is insufficient.

In this work, we begin by asking a fundamental question: **how reliably do biomedical vision-language models handle unanswerable questions, which we define as questions that are unsupported by the accompanying medical image, cannot be resolved from the available visual evidence alone, and should therefore prompt the model to abstain rather than produce a specific answer?** To investigate this, we construct a subset of image-grounded answerable and unanswerable questions from the LLaVA-Med question–answer corpus. To enable abstention from irrelevant or unsupported questions, we design a lightweight answerability pipeline inspired by **CLIP-UP** (Vardi et al., 2026). Given an input question, we first reformulate it into a declarative statement using Qwen (Bai et al., 2023; Ahmed et al., 2025), then jointly encode the statement and its corresponding medical image with BiomedCLIP (Zhang et al., 2023b) to obtain an image–text alignment representation. This representation is used to train a linear layer that learns to distinguish aligned from misaligned image–question pairs. At inference time, the learned linear layer serves as the key answerability module, indicating whether the question is sufficiently grounded in the image and, consequently, whether the model should proceed with answering or abstain.

In a nutshell, we have made 3 contributions to the field of Medical VQA.

- We construct a new **image-grounded medical VQA dataset containing both answerable and unanswerable questions**, designed to evaluate whether medical vision-language models can distinguish valid queries from those unsupported by the image.
- We develop a **lightweight abstention-oriented architecture** that uses question-to-statement reformulation, BiomedCLIP-based image–text alignment, and a learned linear layer to determine whether a question is sufficiently grounded in the medical image before answering.

- We conduct a **cross-domain analysis of answerability and abstention**, evaluating domain-specific and jointly trained projection layers across five medical imaging domains and analyzing the domain structure of BioMedCLIP correlation signals to understand cross-domain generalization.

2 Related Work

2.1 Medical VQA

Medical Visual Question Answering (Med-VQA) has progressed from small, task-specific benchmarks toward increasingly capable generative vision-language models. Early datasets such as VQA-RAD (Lau et al., 2018), PathVQA (Naseem et al., 2022), and SLAKE (Liu et al., 2021) established benchmark settings across radiology and pathology, while domain-specific pretraining approaches such as PubMedCLIP (Eslami et al., 2023) improved biomedical image-text representations for downstream VQA. More recent work shifted Med-VQA toward generative and instruction-following systems, including PMC-VQA/MedVInT (Zhang et al., 2023c), Med-Flamingo (Moor et al., 2023), and LLaVA-Med (Li et al., 2023a), which enabled open-ended biomedical question answering and multimodal conversation. Following LLaVA-Med, several specialized medical VLMs further expanded this direction, including XrayGPT (Thawakar et al., 2024) for chest X-ray interpretation, RaDialog (Pellegrini et al., 2023) for radiology report generation and dialogue, and PathChat (Lu et al., 2024) for conversational pathology assistance. Collectively, these developments reflect the transition of Med-VQA from closed-set prediction toward general-purpose medical assistants capable of free-form interaction across diverse biomedical imaging domains.

2.2 Unanswerable VQA

Unanswerable VQA has evolved from early work on detecting irrelevant or false-premise questions to explicitly teaching multimodal models when to abstain. Toor et al. (2017) studied question-part relevance and editing for context-aware VQA, while Mahendru et al. (2017) explored using question premises to identify invalid queries. Later, Zhang et al. (2023d) formulated VQA as a setting in which models must reject unanswerable questions while answering valid ones and **UNK-VQA** (Guo et al., 2024) further evaluated abstention through carefully perturbed image-question pairs. **VISREAS** (Akter et al., 2024) extended this setting to complex compositional reasoning with both answerable and unanswerable queries, while Miyai et al. (2025) formalized several categories of unsolvable multimodal questions. More recent studies such as **VisionTrap** (Saadat et al., 2025) examined logically unanswerable visual questions, and Hsieh et al. (2025) extended rejection behavior to vision-language-action models faced with impossible instructions. In the medical domain, **Gut-VLM** (Khanal et al., 2025) investigated visually unsupported model outputs, but did not explicitly frame general medical VQA as a question-answerability problem. In light of these recent works and their inherent limitations, to the best of our knowledge, **our work is the first to study image-grounded answerability and abstention in general medical VQA.**

3 Unanswerable Question Generation

We construct a multi-domain medical VQA dataset spanning five clinically distinct imaging modalities: **X-ray, MRI, CT scan, Histology, and Gross Pathology**. Medical images and their associated question-answer pairs are sourced from PMC-15M (Zhang et al., 2023a), following the data pipeline established by LLaVA-Med (Li et al., 2023a).

3.1 Image-Grounded Question Filtering

A key challenge in medical VQA is that not every incoming question requires visual evidence. We initially start from the **LLaVA-Med 60K instruction-following** dataset, but observe that a substantial portion of its questions can be answered without consulting the associated image. Since our downstream answerability module is specifically designed to determine whether a vision-dependent question is supported by the image, applying it to such non-visual questions would be unnecessary and could confound the answerable/unanswerable formulation. We therefore introduce **Guardrail**, a learned *question-routing module* that classifies each question as either visually or non-visually grounded. To construct its training data, questions are first annotated with **KEEP/REMOVE** labels, where non-image-grounded, generic, binary, and general medical-knowledge questions are excluded from the visual answerability pathway. Further details on the annotation criteria and filtering procedure are provided in Appendix **??**. A DistilBERT (Sanh et al., 2019) classifier is then trained on these

labels. At inference time, questions predicted to require visual evidence are forwarded to the subsequent answerability detection pipeline, whereas non-visual questions bypass this stage and are sent directly to the language model for answering. We provide a detailed evaluation of Guardrail in the Appendix B and C, including domain-specific Precision, Recall, F1, and ROC-AUC results and demonstrated the types of question we filtered in Appendix and A and E.

3.2 Statement-Aware CLIP Signal Extraction

Having filtered the dataset to retain questions that genuinely require visual evidence, we next focus on whether the semantic content of each question is supported by the corresponding medical image. A core component of our framework is therefore an image–text alignment signal that captures this relationship. Inspired by CLIP-UP (Vardi et al., 2026), we obtain this signal using a contrastive vision–language encoder. CLIP-UP employs Structure-CLIP (S-CLIP) (Huang et al., 2024), which is designed to better distinguish texts with similar syntactic structures but different semantics. These correlation vectors provide an explicit global alignment prior that is subsequently projected into the VLM feature space.

The direct transfer of this formulation to the biomedical domain, however, introduces an important text-distribution mismatch. We use BioMedCLIP (Zhang et al., 2023b), which was pretrained on approximately 15 million biomedical image–text pairs from PubMed Central, where the textual modality predominantly consists of descriptive figure captions. Consequently, its text encoder is naturally aligned with declarative descriptions of visual content rather than interrogative VQA prompts. For example, expressions such as *What abnormality is visible in the left lung?"*, *Where is the lesion located?"*, or *"What is the appearance of the right costophrenic sinus?"* ask for information rather than explicitly describing the visual concept that should align with the image. Directly embedding such questions may therefore provide a weaker alignment signal than text resembling BioMedCLIP's pretraining distribution.

To reduce this mismatch, we introduce a **question-to-statement transformation** prior to BioMedCLIP encoding. Each visually grounded question is reformulated using Qwen2.5-1.5B-Instruct (Ahmed et al., 2025) into a concise declarative statement while preserving its underlying visual semantics. For instance, "What abnormality is visible in the left lung?" is transformed into *"The image shows an abnormality in the left lung."* Importantly, this transformation does not attempt to answer the question; rather, it converts the interrogative form into a caption-like representation more compatible with BioMedCLIP. Further examples and analysis are provided in Appendix D and F.

The medical image and converted statement are encoded using BioMedCLIP, and their ℓ_2-normalized embeddings are combined via element-wise multiplication to form a 512-dimensional correlation vector. Following CLIP-UP, this vector is projected into the VLM feature space through a learned linear layer and used as an answerability prior. Unlike CLIP-UP, which directly uses S-CLIP, our formulation first converts questions into declarative statements to better match BioMedCLIP's biomedical pretraining distribution.

3.3 Generating Unanswerable Questions

At this stage, each retained sample consists of the medical image, its visually grounded question, and the corresponding declarative statement. The next challenge is to construct reliable unanswerable counterparts for these samples. Synthetic unanswerable questions may introduce linguistic or distributional artifacts that make them distinguishable from answerable questions based on surface form alone, reducing the need for genuine visual reasoning (Rajpurkar et al., 2018; Davis, 2020), while randomly sampled questions may be either trivially irrelevant to the target image or remain semantically compatible with it, resulting in weak negatives or false-negative label noise (Robinson et al., 2020; Jang and Wang, 2023; Li et al., 2023b). We therefore seek already generated questions that are not related to the specific image, hence rendering them as unanswerable.

Let $\mathcal{E} = \{(q_i, s_i, I_i)\}_{i=1}^{N}$ denote the set of filtered question-statement-image triples for a given domain, where q_i is the question, s_i its corresponding declarative statement, and I_i the associated image. For each entry i, we seek a negative statement s_k from a different image such that s_k is semantically incompatible with I_i, forming an unanswerable pair.

We encode all statements using BioMedCLIP's text encoder f_T and all images using its image en-

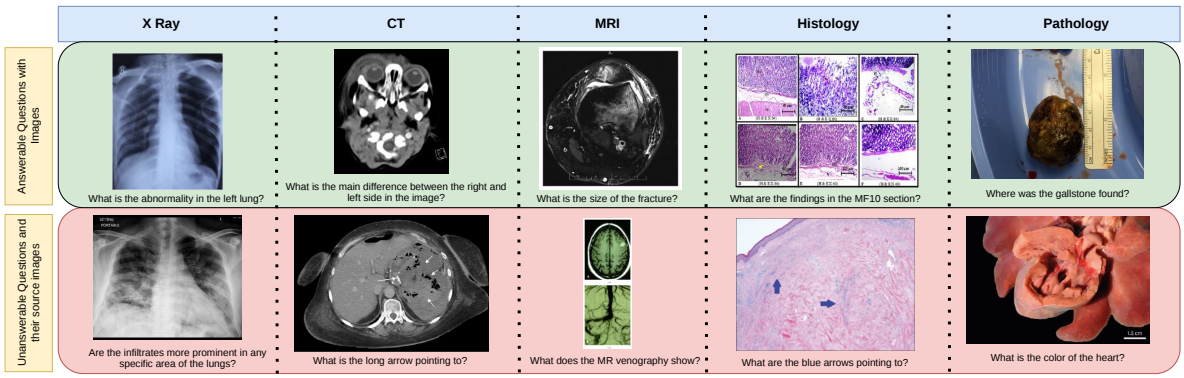


Figure 3: Examples of answerable and constructed unanswerable image-question pairs across five medical imaging domains. The top row shows the original answerable image-question pairs. The bottom row shows the source image-question pairs whose questions are transplanted onto the corresponding top-row images to construct unanswerable examples.

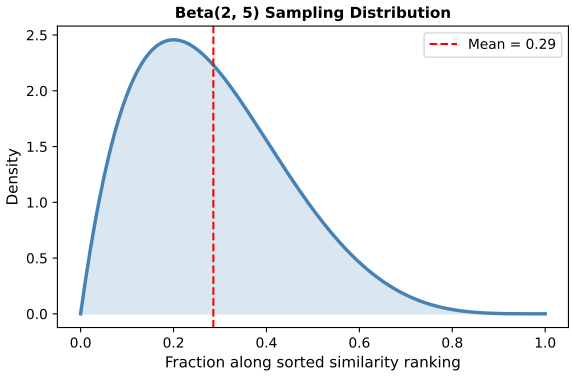


Figure 4: Skewed toward lower similarity values to select semantically related but visually mismatched statements, avoiding both trivial and near-duplicate negatives.

coder f_V, yielding ℓ_2-normalized feature vectors:

$$\mathbf{t}_j = \frac{f_T(s_j)}{\|f_T(s_j)\|_2}, \quad \mathbf{v}_i = \frac{f_V(I_i)}{\|f_V(I_i)\|_2}$$

For image I_i, the cosine similarity with each candidate statement s_j is:

$$\sigma_{ij} = \mathbf{v}_i^\top \mathbf{t}_j, \quad j \notin \mathcal{S}_i$$

where $\mathcal{S}_i = \{j : I_j = I_i\}$ is the set of indices sharing the same image, excluded to prevent false positives. The valid candidates are then ranked in ascending order of similarity, producing an ordered index set π_i such that $\sigma_{i,\pi_i(1)} \leq \sigma_{i,\pi_i(2)} \leq \cdots \leq \sigma_{i,\pi_i(M)}$, where $M = N - |\mathcal{S}_i|$.

The negative is selected by sampling a fractional rank $\alpha \sim \text{Beta}(2, 5)$ and picking:

$$k^* = \pi_i\left(\lfloor \alpha \cdot (M-1) \rfloor + 1\right)$$

The Beta$(2, 5)$ distribution has mean $\mathbb{E}[\alpha] = \frac{2}{7} \approx 0.29$, concentrating selections toward the lower similarity quartile, as illustrated in Fig. 4. This ensures that the negatives are semantically incompatible with I_i while avoiding the degenerate case of always selecting the globally least-similar statement, which would produce trivially detectable negatives, as shown in Appendix G, H and I. The resulting unanswerable record pairs I_i with the question q_{k^*} and statement s_{k^*} from a different image, with ground-truth answer set to *"I cannot answer."*

Fig. 3 illustrates representative answerable and unanswerable image–question pairs across five medical imaging domains. The top row shows the original answerable image–question pairs, while the bottom row shows the source image–question pairs whose questions were transplanted onto the top-row images to create the unanswerable examples.

Table 1 summarizes the dataset statistics across all five domains, comprising 36,750 training and 9,192 test samples in total. CT and MRI are the largest domains, while Gross Pathology is the smallest due to its limited representation in the source data. All splits are performed at the image level to prevent leakage, and answerable and unanswerable pairs are perfectly balanced by construction across every domain and split.

4 Training Details

We train a lightweight linear projection layer on top of the frozen BioMedCLIP signal as the abstention classifier. The layer is a single fully-connected projection with no bias and no non-linearity, mapping the 512-dimensional BioMedCLIP image–text sim-

Table 1: Dataset statistics for each medical imaging domain. Each split contains an equal number of answerable and unanswerable question-image pairs. The validation set serves as the test set in all experiments.

	Train		Test	
Domain	**Answerable**	**Unanswerable**	**Answerable**	**Unanswerable**
X-ray	1,539	1,539	385	385
MRI	4,960	4,960	1,241	1,241
CT	6,400	6,400	1,601	1,601
Histology	4,916	4,916	1,229	1,229
Gross Pathology	560	560	140	140
Total	**18,375**	**18,375**	**4,596**	**4,596**

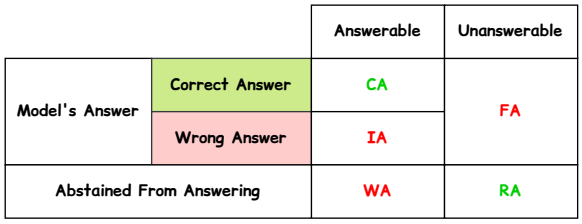


Figure 5: Evaluation outcomes for answerable and unanswerable questions, covering correct answering, incorrect answering, and abstention behavior.

ilarity embedding to the 4096-dimensional hidden space of the LLaVA-Med language model (2.1M trainable parameters). The simplicity is deliberate: a linear layer is sufficient to exploit the separability already present in the CLIP signal between matched and mismatched image–text pairs.

Six variants are trained in total: five domain-specific models (chest X-ray, MRI, CT, histology, gross pathology) and one combined model trained jointly across all five domains. The domain-specific models establish in-domain abstention performance, while the combined model examines whether abstention behavior transfers across domains. All models are trained on a single NVIDIA A100 GPU for three epochs.

We train only a 2.1M-parameter linear projection rather than LoRA, which at the standard rank-128 configuration trains 356.5M parameters (4.5% of LLaVA-Med), about 170× more, and modifies the language model itself. Keeping LLaVA-Med frozen preserves its original behaviour and lets the abstention module be removed or swapped per domain. Vardi et al. (2026) find embedding injection competitive with LoRA in the general domain; we leave a medical-domain comparison to future work.

5 Evaluation metrics

Building on the confusion-matrix-style evaluation frameworks introduced by Madhusudhan et al. (2025) and Saadat et al. (2026), we categorize model behavior according to whether the input question is answerable or unanswerable and whether the model responds or abstains. As illustrated in Fig. 5, an answerable question can result in a **Correct Answer (CA)** when the model responds correctly, an **Incorrect Answer (IA)** when it responds incorrectly, or a **Wrong Abstention (WA)** when it unnecessarily refuses to answer.

For an unanswerable question, any generated response is treated as a **Failure to Abstain (FA)**, whereas correctly withholding an answer is categorized as a **Right Abstention (RA)**. In our analysis, we primarily focus on CA, WA, and RA to capture the trade-off between answer accuracy and appropriate abstention.

Classifying responses into **CA, WA and RA** requires judging correctness, which in open-ended medical VQA cannot rely on lexical overlap, since a correct response may be worded differently from the reference. We therefore use GPT-5.4 (OpenAI, 2022) as an LLM judge, given the reference answer and question context, following the growing use of LLM-based evaluators for free-form generation (Li et al., 2025a; Gu et al., 2026; Fu et al., 2024; Liu et al., 2023b; Zheng et al., 2023). LLaVA-Med (Li et al., 2023a) similarly used GPT-4 to evaluate biomedical responses.

6 Results

Table 2 reports CA, WA, and RA for all domain-specific LLaVA-Med variants, the Combined model, and the original LLaVA-Med across the five medical domains. We additionally evaluate the same metrics for several general-purpose VLMs to provide a broader comparison of answering and abstention behavior.

6.1 Abstention Trade-off Analysis

Because our framework does not employ an explicit abstention threshold, we analyze RA and WA jointly at the model's natural operating point. The Combined model achieves substantial gains in RA over the original LLaVA-Med, increasing from 12.7 to 87.5 on X-ray, 10.6 to 93.3 on CT, 8.5 to 93.9 on MRI, and 8.7 to 92.9 on Histology. However, these gains are accompanied by higher WA in several domains; for example, WA increases from 2.1 to 19.2 on X-ray and from 2.8 to 17.6 on Histology. *This indicates that improved rejection of unsupported questions can also increase unnecessary abstention on answerable inputs, motivating joint*

Model/Training Setup	X-ray			CT			MRI			Histology			Gross Pathology		
	CA ↑	WA ↓	RA ↑	CA ↑	WA ↓	RA ↑	CA ↑	WA ↓	RA ↑	CA ↑	WA ↓	RA ↑	CA ↑	WA ↓	RA ↑
LLaVA-Med															
LLaVA Med	40.0	2.1	12.7	28.1	1.5	10.6	32.2	3.0	8.5	28.3	2.8	8.7	30.7	1.4	14.3
LLaVA Med + X-ray	31.7	17.1	87.3	29.3	2.1	24.7	29.7	6.8	42.2	25.2	8.0	36.5	30.0	0.0	12.1
LLaVA Med + CT	39.7	1.0	4.7	26.2	13.4	90.9	30.3	3.8	37.1	26.4	7.2	46.3	30.0	0.7	10.7
LLaVA Med + MRI	42.6	1.3	4.9	30.4	2.2	32.2	27.5	12.5	95.3	26.5	6.1	35.6	33.6	2.9	21.4
LLaVA Med + Histology	41.3	0.3	2.1	30.0	0.9	13.1	28.8	1.9	22.6	25.6	11.8	92.4	30.0	0.0	13.6
LLaVA Med + Pathology	41.8	0.8	3.1	27.6	2.2	22.0	29.7	3.3	23.0	26.4	5.4	32.7	28.6	10.7	82.1
LLaVA Med + Combined	36.1	19.2	87.5	26.2	13.2	93.3	27.9	12.0	93.9	22.1	17.6	92.9	32.1	7.1	69.3
General-Purpose VLMs															
GPT-5.4 nano	46.8	2.6	15.1	28.7	8.2	17.0	43.5	9.1	23.9	36.0	3.8	15.9	32.1	3.6	12.9
GPT-5.4 mini	62.3	0.5	13.5	45.0	7.1	34.4	51.2	7.4	30.6	48.5	1.4	15.3	40.7	2.1	10.7
Gemini 3.5 flash lite	59.2	1.3	20.8	46.3	7.2	45.0	51.1	6.0	49.2	48.6	2.5	33.4	37.1	7.9	31.4
Gemini 3.8 flash	64.4	0.3	15.8	56.4	1.3	39.2	60.0	1.9	36.1	56.8	1.5	25.2	50.7	2.1	23.6
Claude Haiku 4.5	37.1	3.9	13.0	25.2	9.3	18.4	32.3	11.1	27.2	32.4	5.1	15.5	25.7	10.7	17.9

Table 2: Cross-domain evaluation of domain-specific LLaVA-Med variants and general-purpose VLMs. CA, WA, and RA denote Correct Answer, Wrong Abstention, and Right Abstention, respectively. Green and red cells indicate the best and worst performance within each model group, respectively.

consideration of RA and WA rather than evaluating abstention performance through RA alone.

6.2 Domain Specificity of CLIP Signals and its Implications for Cross-Domain Generalization

Projection layers trained on a single imaging domain transfer poorly to other domains. To understand why, we analyze the 512-dimensional BioMedCLIP signal, the element-wise product of the ℓ_2-normalized image and text embeddings. Since BioMedCLIP and the language model are frozen, the projection layer is the only trainable component and learns only from the signal distribution seen in training. As shown in Fig. 6, t-SNE projections of both answerable and unanswerable samples form clear domain-specific clusters across X-ray, CT, MRI, Histology, and Gross Pathology, indicating substantially different signal distributions.

This separation helps explain the limited cross-domain transfer of domain-specific projection layers. A layer trained on one modality is calibrated to a particular region of the CLIP signal space and may therefore behave unreliably when applied to signals from a different domain. In contrast, combined training exposes the projection layer to signals from all five modalities, allowing it to learn a mapping over a broader range of domain-specific distributions. *This observation motivates the use of domain-specific layers when the deployment domain is known, while a jointly trained layer provides a more suitable choice for general-purpose medical VQA settings.*

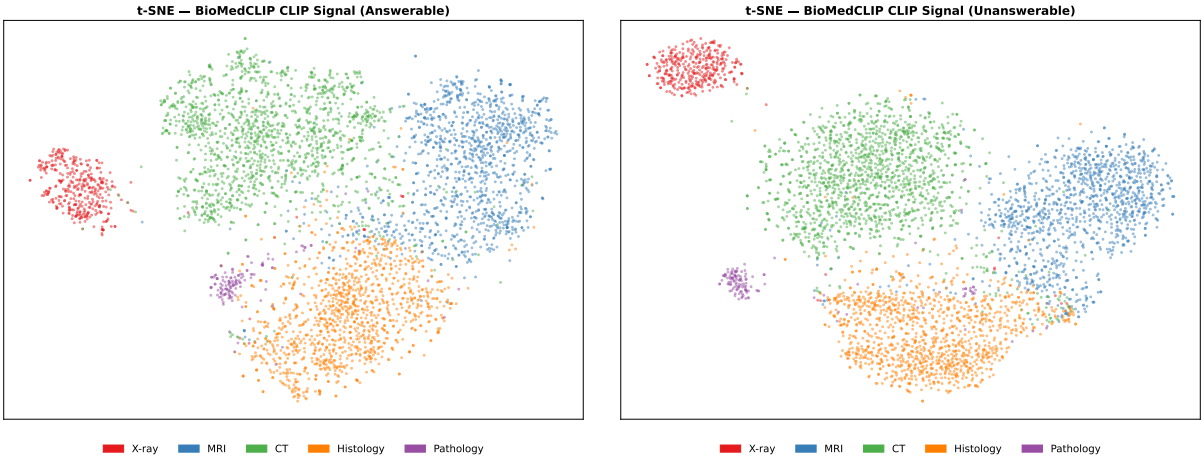
6.3 General-Purpose VLMs vs. Adapted LLaVA-Med

Frontier general-purpose VLMs such as GPT-5 perform strongly on medical VQA and clinical reasoning benchmarks, sometimes outperforming specialized medical VLMs (Buckley et al., 2026; Wang et al., 2026; Hu et al., 2026; Li et al., 2025b; Yan et al., 2023). We test whether this extends to answerability awareness. As shown in Tab. 2, *these models generally achieve higher Correct Answer (CA) scores than the LLaVA-Med variants*, with Gemini 3.8 Flash reaching 50.7–64.4% across domains. However, *their Right Abstention (RA) scores remain substantially lower than those of the adapted LLaVA-Med variants*, peaking at 49.2%. Thus, *strong answering ability does not imply awareness of whether a question is supported by the image.*

6.4 Analysis of Model Failure Patterns on Unanswerable Questions

To investigate abstention behavior across heterogeneous medical imaging domains, we focus on the CLIP-UP combined model, which was trained jointly on all five domains.

Of the 4,596 unanswerable questions, the model failed to abstain on 362 (7.9%). Fig. 7 illustrates the lexical distribution of these failures, decomposed by domain. Two dominant failure categories



(a) Answerable (b) Unanswerable

Figure 6: t-SNE visualization of BioMedCLIP CLIP signals colored by domain for (a) answerable and (b) unanswerable question-image pairs. Both plots reveal well-separated domain clusters, confirming that the CLIP signal space is strongly domain-dependent. This separation explains why models trained on a single domain fail to generalize across domains, as the embedding distributions for each medical modality occupy distinct regions of the feature space.

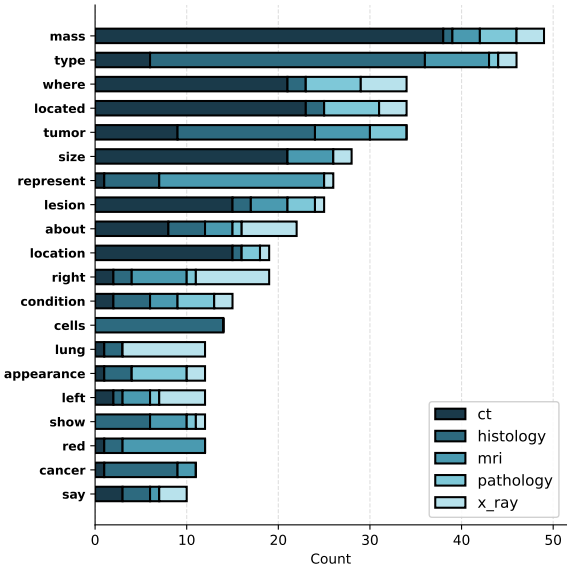


Figure 7: Lexical distribution of failure-to-abstain cases for the Combined model across five medical imaging domains. Bars show the frequency of the most common content words, decomposed by domain.

emerge. The first is anatomy and pathology identification: words such as ‘mass’ (49), ‘tumor’ (34), and ‘lesion’ (25) are the most frequent, suggesting the model hallucinates responses when questions reference specific pathological findings. The second is spatial localization: terms such as ‘where’ (34), ‘located’ (34), and ‘right/left’ indicate a tendency to produce positional answers even when the queried structure is absent from the image.

Domain-level decomposition reveals that CT and MRI account for the majority of failures, while pathology contributes disproportionately to anatomy-specific failures and chest X-ray skews toward localization errors. These findings motivate future work on strengthening the CLIP signal for fine-grained anatomical discrimination, particularly under domain shift.

7 Conclusion

We identify a key limitation of LLaVA-Med: it is not trained to abstain when a question is unsupported by the image. We construct a multi-domain medical VQA dataset of answerable and unanswerable examples and introduce a lightweight BioMedCLIP-based projection layer that adds abstention without retraining the base model. *This parameter-efficient adaptation improves answerability awareness, though with some increase in wrong abstention, and reveals strong domain dependence that joint training partly overcomes. General-purpose VLMs achieve higher answer accuracy yet abstain less reliably, showing that answering capability and answerability awareness are distinct properties.* Overall, our findings show that lightweight alignment-based mechanisms can extend existing medical VLMs with explicit abstention behavior. Future work should strengthen the alignment signal for fine-grained anatomical and localization queries, where most remaining failures occur.

Limitations

Our unanswerable pairs are constructed automatically by transplanting questions between images, and we have not manually verified every pair. We inspected a random subset and found the large majority to be genuinely unanswerable, though a small number of transplanted questions may remain answerable from the new image, for example when both images contain a similar finding such as a mass or lesion. Several stages of our pipeline also rely on large language models, including GPT-5.4 for question filtering and correctness judgment and Qwen2.5 for question-to-statement conversion, so individual decisions may vary with different prompts or model versions. Our aim, however, is not to report absolute scores but to compare answering and abstention behavior; since all models are evaluated with the same prompts and judge, the relative differences in CA, WA, and RA remain consistent. Finally, all images are derived from PubMed Central figures, which may differ from raw clinical scans, and we evaluate on LLaVA-Med only, although the projection layer is model-agnostic and could be applied to other medical VLMs.

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A Excluded Question Types in LLaVA-Med Instruct 60K

The LLaVA-Med Instruct 60K dataset (Li et al., 2023a) contains a diverse mix of medical visual question answering pairs, not all of which require genuine visual grounding. Prior to constructing our Unanswerable counterparts, we filter out four categories of questions that are unsuitable for the visual answerability pathway: non-image-grounded questions, generic questions, binary questions, and general medical knowledge questions. Table 3 provides representative examples of each type. We use GPT-5.4 to assign the initial **KEEP/REMOVE** labels according to these filtering criteria, which are then used to train the DistilBERT-based guardrail classifier.

Non-image-grounded questions are posed without an accompanying image reference, or are answerable entirely from surrounding textual context. For example, *"What happened to the cyst?"* can be resolved from a prior report rather than visual inspection. Including such questions would conflate text-based reasoning with the image-grounded abstention task.

Generic questions are overly broad and lack specificity to a visual finding. Questions such as *"What is the main focus of the image?"* do not test whether a model can identify a particular structure or pathology; they merely ask the model to describe whatever happens to be salient. These questions offer no meaningful discriminative signal for unanswerable detection.

Binary questions elicit a 'yes' or 'no' response and do not require the model to localize, classify, or describe a finding. A question such as *"Are there any differences between the images from day 1 and day 11?"* can be answered affirmatively with minimal visual engagement. Binary responses also complicate GPT-judge evaluation, as a one-word answer is ambiguous between a genuine finding and a refusal.

General medical knowledge questions are definitional or procedural in nature and are answerable without any image at all. Questions such as *Can you describe the axial plane?"* or *What is the purpose of immunohistochemistry?"* test recall of domain knowledge rather than visual reasoning. Retaining them would inflate correct-answer rates independently of image content.

By removing these four question types, the remaining question set constitutes a genuine visual reasoning benchmark in which every question requires direct inspection of the accompanying image to answer correctly.

B Question Quality Filtering via DistilBERT

To filter low-quality questions from our multi-domain medical VQA dataset, we train a binary text classifier based on DistilBERT (Sanh et al., 2019) for each imaging domain. The classifier takes a question string as input and predicts one of two labels: KEEP or REMOVE, according to the filtering criteria described in Section 3.1. We use distilbert-base-uncased as our backbone, a distilled variant of BERT (Devlin et al., 2019) that retains 97% of BERT's language understanding performance at 40% fewer parameters. A linear classification head is appended on top of the [CLS] token representation, producing logits over two classes. The model is fine-tuned end-to-end using the AdamW optimizer with a learning rate of 2×10^{-5} for three epochs, with a batch size of 32.

A significant challenge in training this classifier is **class imbalance**. Across all domains, the majority of questions are labeled REMOVE by the GPT-5.4 annotator, with KEEP labels comprising a smaller fraction. Training a classifier naively under this imbalance causes it to be biased toward the majority class, giving high accuracy but poor recall on KEEP predictions. To mitigate this, we apply class-weighted cross-entropy loss, where the weight assigned to the REMOVE class is 1.0 and

Type	Example Question	Reason for Exclusion
Non-image-grounded	"What is the purpose of immunohistochemistry?"	No visual inspection required; answerable from text alone.
Non-image-grounded	"What happened to the cyst?"	Answered from prior context, not image content.
Generic / Vague	"What is the main focus of the image?"	Too open-ended; not tied to a specific visual finding.
Generic / Vague	"What is the color of the tumor?"	Answerable from any image without domain knowledge.
Binary	"Are there any differences between the CT and MRI scans?"	Yes/no response; does not require visual grounding.
Binary	"Are there any differences between the images from day 1 and day 11?"	Comparison prompt answered with yes/no, not a finding.
General Medical Knowledge	"Can you describe the axial plane?"	Anatomical definition; no image needed.
General Medical Knowledge	"What is the purpose of immunohistochemistry?"	Technique definition; purely textual knowledge.

Table 3: Examples of question types excluded from the visual answerability pathway in LLaVA-Med Instruct 60K.

the weight for KEEP is computed as the ratio of REMOVE to KEEP samples in the training set:

$$w_{\text{KEEP}} = \frac{N_{\text{REMOVE}}}{N_{\text{KEEP}}}$$

This penalizes the model more heavily for misclassifying KEEP instances, encouraging it to learn a more balanced decision boundary.

Since DistilBERT produces a continuous probability score rather than a hard label, a classification threshold must be selected to convert these scores into KEEP/REMOVE decisions. Rather than defaulting to 0.5, which is suboptimal under class imbalance, we select the threshold that maximizes the F1 score on the validation set. Specifically, for each domain we compute the Precision-Recall curve and evaluate F1 at every candidate threshold, selecting:

$$t^* = \arg\max_t \; F_1(t)$$

This threshold is then saved alongside the model checkpoint and applied at inference time. The selected thresholds vary across domains, reflecting differences in class distribution and question style per modality.

We evaluate each domain-specific classifier on a held-out validation set constructed via an 80/20 stratified split. Fig. 9 shows the ROC curves for all six classifiers (five domain-specific and one combined). All models achieve strong discrimination, with AUC scores ranging from 0.960 (Gross Pathology) to 0.991 (CT), demonstrating that question quality is largely predictable from text alone without access to the image. Figure 8 and Tab. 4 present the Precision, Recall, and F1 as a function of threshold for each domain, with the selected optimal threshold marked in red. The consistently high F1 values at the chosen thresholds confirm that the weighted loss and threshold selection together effectively address the class imbalance, yielding a reliable filtering stage across all imaging modalities.

Table 4: Per-domain classification performance of Guardrail (question quality filter) on the held-out validation set. Metrics are reported separately for the **Remove** and **Keep** classes. The threshold t^* is selected per domain by maximising the F1 score on the KEEP class.

Domain	Class	Prec.	Rec.	F1	Support	t^*
X-ray	Remove	0.98	0.99	0.98	3,746	0.777
	Keep	0.89	0.84	0.86	432	
Gross Pathology	Remove	0.97	0.94	0.95	1,236	0.793
	Keep	0.63	0.80	0.71	169	
CT	Remove	0.98	0.97	0.98	7,766	0.844
	Keep	0.87	0.92	0.90	1,748	
Histology	Remove	0.98	0.96	0.97	7,932	0.777
	Keep	0.81	0.90	0.85	1,484	
MRI	Remove	0.97	0.95	0.96	7,661	0.714
	Keep	0.78	0.85	0.81	1,436	
Combined	Remove	0.97	0.94	0.96	28,340	0.619
	Keep	0.72	0.86	0.78	5,267	

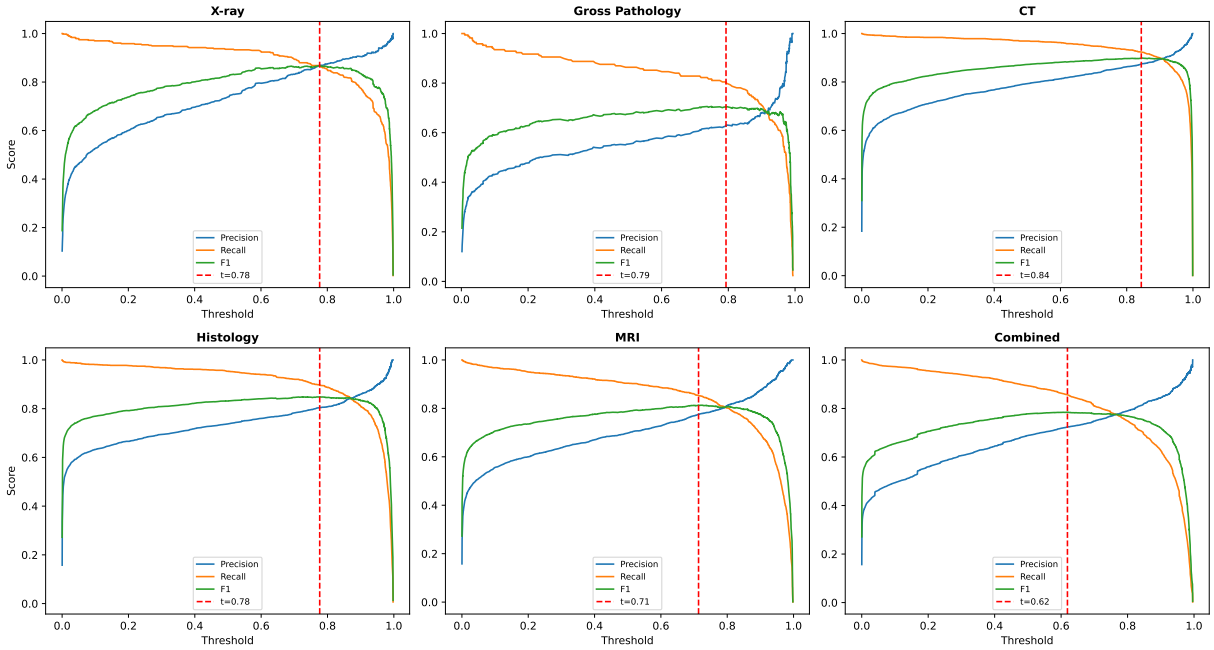


Figure 8: Per-domain curves showing the trade-off between precision and recall across all candidate decision thresholds. The optimal threshold (red dashed line) is selected independently for each domain as the point of maximum F1, accounting for class imbalance between KEEP and REMOVE labels. High F1 values at the selected thresholds confirm that the weighted loss formulation and threshold tuning together yield a reliable filtering stage across all five imaging modalities and the combined setting.

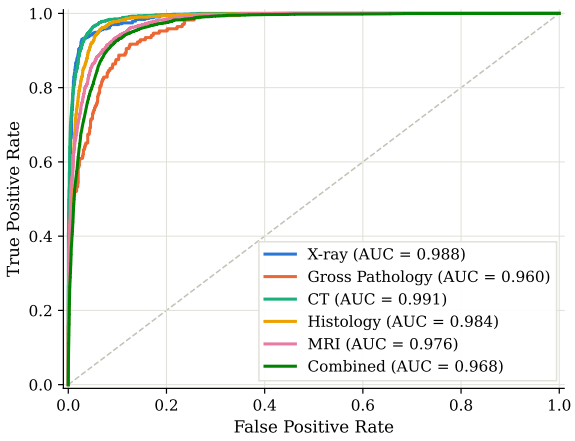


Figure 9: ROC curves for the DistilBERT-based question quality classifier trained independently on each of five imaging domains and on the combined dataset.

C Visual Grounding Filtering

The LLaVA-Med instruction dataset (Li et al., 2023a) contains 168,032 question-answer pairs across five medical imaging domains. However, a substantial proportion of these questions do not require visual grounding, they can be answered from general medical knowledge, are binary yes/no questions, or are so generic that they apply to virtually any image in the domain. Including such questions in training would introduce noise and undermine the model's ability to learn meaningful image-question alignment signals.

To address this, we apply a domain-specific filtering step using GPT to classify each question as KEEP or REMOVE based on four criteria: (1) the question must be visually grounded and answerable solely from the image; (2) it must be open-ended and not answerable with a simple yes or no; (3) it must be specific enough to require the particular image rather than any image from the domain; and (4) it must not ask for general medical knowledge or textbook definitions. Table 5 reports the number of questions retained after filtering for each domain.

The filtering is aggressive by design, retaining only 13.7% of the original questions on average. This reflects the high proportion of clinically framed but visually ungrounded questions in the source dataset. The retained questions form a high-quality subset that genuinely requires visual reasoning, providing a sound foundation for constructing the unanswerable question pairs described in Section 3.3.

Table 5: Effect of visual grounding filtering on the raw LLaVA-Med instruction dataset. Only questions requiring genuine visual grounding on the specific image are retained.

Domain	Raw	Filtered	Retention (%)
X-ray	20,886	1,924	9.2
MRI	45,481	6,201	13.6
CT	47,568	8,001	16.8
Histology	47,076	6,145	13.1
Gross Pathology	7,021	700	10.0
Total	**168,032**	**22,971**	**13.7**

D Question-to-Statement Conversion

To extract a discriminative BioMedCLIP signal, each VQA question is converted into a declarative statement before being passed to the text encoder. The conversion follows a simple domain-prefixed template: for example, *What is the tracheal shadow like?"* becomes *The chest X-ray shows the tracheal shadow."* Table 6 provides representative examples of this conversion across all five imaging domains.

E Distribution of Question Types

We analyze the 18,375 standard (answerable) questions across all five medical domains. Figure 10 shows the most frequent content words and their co-occurrence structure. The questions are dominated by four broad categories: classification (e.g., *type*, *represent*), pathology identification (*mass*, *tumor*, *lesion*), localization (*where*, *located*, *left*, *right*), and measurement/comparison (*size*, *between*). These patterns indicate that the retained questions primarily require image-grounded reasoning rather than generic medical knowledge.

F Effect of Question-to-Statement Reformulation

BioMedCLIP is pretrained predominantly on biomedical image–caption pairs, whereas medical VQA inputs are naturally expressed as interrogative questions. To examine whether reformulating questions into declarative statements better matches the encoder's pretraining distribution, we compare BioMedCLIP image–text cosine similarity using the original question and its Qwen-generated statement counterpart on the answerable X-ray validation set.

Using the original questions, the average image–text similarity is 0.2592 ± 0.0709, while the corresponding declarative statements achieve a substantially higher similarity of 0.3758 ± 0.0652. The statement representation yields higher similarity in 356 out of 385 samples (92.5%), indicating that the reformulation consistently improves alignment with the corresponding medical image.

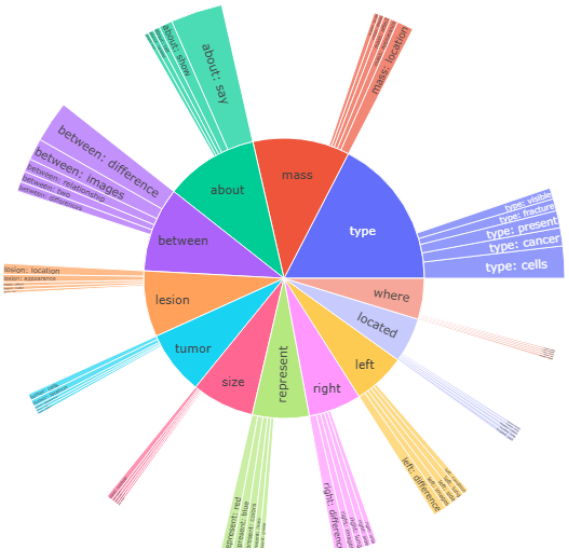


Figure 10: Sunburst visualization of dominant lexical patterns in the standard answerable questions.

We further compare statement-based similarities for answerable and unanswerable image–question pairs. Answerable pairs obtain an average similarity of 0.3758 ± 0.0652, whereas unanswerable pairs obtain 0.2999 ± 0.0596. This separation suggests that the statement formulation not only produces stronger alignment with BioMedCLIP, but also preserves a meaningful distinction between visually supported and unsupported questions. These results support the use of question-to-statement reformulation prior to BioMedCLIP encoding in our answerability pipeline.

G Negative Sampling Strategy: Beta Sampling vs. Lowest Similarity

A critical design choice in constructing the unanswerable dataset is the strategy used to select negative statements, statements that are semantically incompatible with the query image and thus render the associated question unanswerable. In this section, we ablate two candidate strategies and demonstrate that the choice of sampling method has a profound effect on dataset quality.

Lowest Similarity Sampling. The most intuitive approach is to always select the statement with the lowest cosine similarity to the image under BioMedCLIP's joint embedding space, that is, to

Domain	Question	Statement
Chest X-ray	What is the tracheal shadow like?	The chest X-ray shows the tracheal shadow.
	What do the infiltrates look like?	The chest X-ray shows infiltrates.
	Where is the mass located?	The chest X-ray shows a mass.
	What is the position of the IABP in the image?	The chest X-ray shows the position of the Intra-Atrial Balloon Pump (IABP).
	What are the findings in the right inferior lobe?	The chest X-ray shows findings in the right inferior lobe.
MRI	What types of brain tumors are shown in the PET images?	The MRI shows various types of brain tumors.
	What type of lesion is shown in the image?	The MRI shows a lesion.
	Is this image taken before or after surgery?	The MRI shows the imaging sequence taken prior to surgery.
	What do the lower four rows represent?	The MRI shows the lower four rows representing the lumbar spine region.
CT	What abnormalities are seen in the right upper lobe?	The CT demonstrates abnormalities in the right upper lobe.
	What type of tumor is shown in the image?	The CT shows a tumor in the image.
	What is the finding on the distal gastric wall?	The CT shows a finding on the distal gastric wall.
	What is the location of the mass?	The CT shows the mass located in the chest.
Histology	What type of cells are infiltrating the bone marrow?	The histology slide shows cells infiltrating the bone marrow.
	What is the lining of the cyst?	The histology slide shows the histopathological appearance of the lining of the cyst.
	What is the region of the fallopian tube shown?	The histology slide shows the region of the fallopian tube.
	What can you say about the neutrophils in the different groups?	The histology slide demonstrates the characteristics of neutrophils across various groups.
Gross Pathology	What is the main feature in the ablation zone?	The gross specimen shows the main feature in the ablation zone.
	What can you say about the thickness of the gastric wall?	The gross specimen shows the thickness of the gastric wall.
	What is the difference between the left and right tibia?	The gross specimen shows the difference between the left and right tibia.
	What is the consistency of the tumor?	The gross specimen shows the consistency of the tumor.

Table 6: Examples of question-to-statement conversion across five medical imaging domains. Each statement is used as the text input to BioMedCLIP for CLIP signal extraction.

set $\alpha = 0$ and always pick $k^* = \pi_i(1)$. This guarantees maximal visual incompatibility between the image and the negative statement. However, this strategy has an unintended consequence: the globally least similar statement to a given medical image tends to originate from a visually and semantically distant context, producing questions that are lexically distinctive from the answerable questions in the dataset. A text-only classifier, one that reads only the question string with no access to the image, can therefore learn to distinguish answerable from unanswerable questions purely on surface-level linguistic cues, bypassing any visual reasoning entirely.

To quantify this, we train a TF-IDF + Logistic Regression classifier on question text alone for each domain and report classification accuracy. As shown in Fig. 11, the lowest-similarity strategy yields text-only accuracies of 89–96% across all five domains, far above the random-chance baseline of 50%. This confirms that the resulting dataset contains substantial *textual bias*, the unanswerable questions are detectable from text alone, rendering the visual grounding requirement trivial for any model that picks up on these cues.

Beta(2,5) Sampling. To address this, we adopt a stochastic sampling strategy governed by a $\text{Beta}(2, 5)$ distribution over the similarity-ranked candidate list. Rather than always selecting the globally least similar statement, we sample a fractional rank $\alpha \sim \text{Beta}(2, 5)$ and select the statement at position $\lfloor \alpha \cdot (M+1) \rfloor$ in the ascending similarity ranking. With a mean of $\mathbb{E}[\alpha] \approx 0.29$, this distribution concentrates selections in the lower quartile

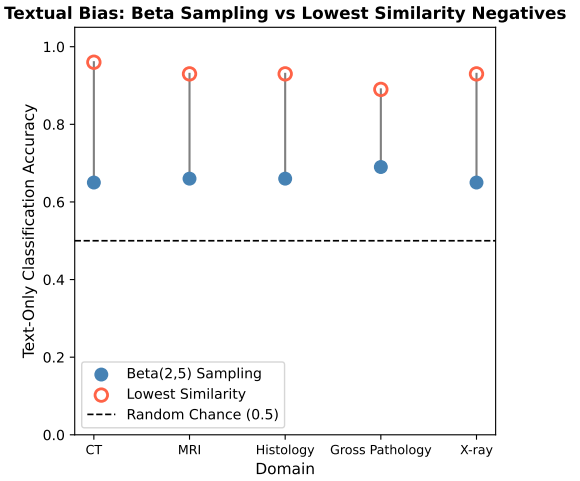


Figure 11: Text-only classification accuracy for unanswerable questions generated using Beta(2,5) sampling versus lowest-similarity negative selection across five medical domains.

of the similarity spectrum while introducing controlled stochasticity in the difficulty of the selected negative. Crucially, the sampled negatives remain close enough in embedding space to the image that their associated questions are lexically similar to answerable questions, they ask about real visual findings from the same domain, just findings that do not appear in the query image.

The same text-only classifier applied to the Beta-sampled dataset yields accuracies of only 65–69% across all domains, barely above random chance. This confirms that our dataset does not exhibit exploitable textual bias: a model cannot determine answerability from question text alone and must leverage the image-question alignment signal to succeed.

This comparison, summarized in Fig. 11, provides strong empirical justification for the Beta sampling design. The dumbbell plot illustrates the gap between the two strategies across all five domains, the hollow markers (lowest similarity) consistently reach near-perfect text-only accuracy, while the solid markers (Beta sampling) remain close to the random-chance line, confirming that our negative construction strategy is vision-dependent by design.

H Visualizing Answerable vs. Unanswerable CLIP Signals

To further validate the quality of our negative sampling strategy, we visualize the BioMedCLIP CLIP signals for answerable and unanswerable question pairs across all five medical imaging domains. For each record in the validation set, we extract the 512-dimensional CLIP signal, the element-wise product of the ℓ_2-normalized image and statement feature vectors, and project it to two dimensions using t-SNE (Van der Maaten and Hinton, 2008).

Fig. 12 presents the resulting embeddings for each domain separately, with answerable records shown in blue and unanswerable records in red. Across all five domains, X-ray, MRI, CT, Histology, and Gross Pathology, the two classes exhibit substantial overlap in the projected embedding space, with no clear boundary separating answerable from unanswerable signals. This is a direct consequence of our Beta(2,5) sampling strategy: the selected negative statements are drawn from the lower similarity quartile but not the absolute minimum, producing CLIP signals that are visually plausible and thus geometrically proximate to the answerable signals in the embedding space.

This observation has two important implications. First, it confirms that the unanswerable questions cannot be trivially rejected based on their CLIP signal alone, the raw BioMedCLIP representation does not distinguish the two classes. Second, it motivates the need for a learned projection, the CLIP-UP linear layer, which is trained to map these overlapping signals into a space where answerability can be reliably detected. The difficulty of this separation, as evidenced by the t-SNE plots, directly reflects the challenge that CLIP-UP is designed to address.

I Inter-Domain Similarity of Unanswerable CLIP Signals

To further quantify the domain specificity of unanswerable CLIP signals, we compute the centroid of each domain's unanswerable signal distribution as $\mu_d = \frac{1}{N_d} \sum_i \mathbf{z}_i$, where $\mathbf{z}i$ are the 512-dimensional CLIP signals for all unanswerable validation records in domain d. We then compute the pairwise cosine similarity $Sij = \cos(\mu_i, \mu_j)$ between all domain centroids and visualize the result as a 5×5 heatmap in Fig. 13.

The off-diagonal similarities range from 0.649 to 0.864, confirming that while all domains share a common medical imaging context, their unanswerable signal distributions occupy meaningfully different regions of the BioMedCLIP embedding space. Gross Pathology exhibits the lowest inter-domain similarity overall (0.649–0.770), consis-

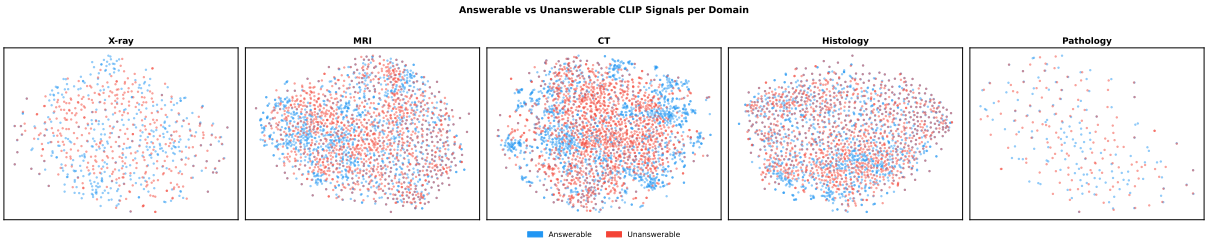


Figure 12: t-SNE visualization of BioMedCLIP CLIP signals for answerable (blue) and unanswerable (red) question-image pairs across five medical imaging domains. Each point represents the 512-dimensional element-wise product of the ℓ_2-normalized image and statement feature vectors, projected to two dimensions. The substantial overlap between classes across all domains confirms that the raw BioMedCLIP signal cannot distinguish answerable from unanswerable questions, motivating the need for the learned CLIP-UP projection layer.

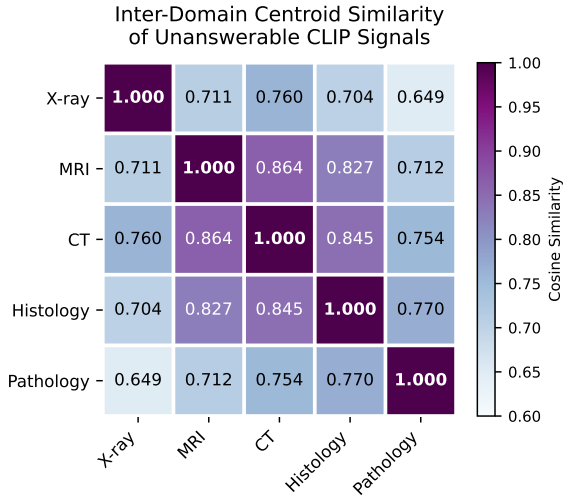


Figure 13: Inter-domain cosine similarity between centroids of unanswerable BioMedCLIP signal distributions across five medical imaging domains.

tent with its visually distinct appearance relative to radiological modalities. MRI, CT, and Histology form a tighter cluster (0.827–0.864), reflecting their shared structural characteristics. These centroid distances directly explain why a projection layer trained on one domain's CLIP signals fails to generalize to another, the signal distribution it was optimized for is geometrically offset from the target domain's distribution, resulting in degraded answerability detection at test time.